Program 1: Banana Level – Student Academic Summary

Problem Statement:

A university wants to generate a quick academic summary for a student.

Write a C program that reads Student Roll Number and Marks obtained in **five subjects**

The program should calculate and display: Total marks and Average marks

INPUT :

```c
#include<stdio.h>
void main()
{
int r;
float m1,m2,m3,m4,m5,tm,avg;
printf("Enter the roll no of student and marks in 5 subjects\n");
scanf("%d %f %f %f %f %f",&r,&m1,&m2,&m3,&m4,&m5);
tm=m1+m2+m3+m4+m5;
avg=tm/5;
printf("total marks : %f\naverage marks : %f",tm,avg);
}
```

OUTPUT :

Output

```
Enter the roll no of student and marks in 5 subjects
2 89 98 87 77 90
total marks : 441.000000
average marks : 88.199997
```

Program 2: Banana Level – Monthly Electricity Bill Estimation

Problem Statement:

An electricity board estimates the monthly bill based on consumption.

Write a C program that reads Number of units consumed and Cost per unit

The program should compute and display: Total electricity bill amount

INPUT :

```c
#include <stdio.h>
#define cost 7
void main()
{
int u,t;
printf("Enter number of units ");
scanf("%d", &u);
t=cost*u;
printf("Total Bill Amount %d",t);
}
```

OUTPUT :

Output

```
Enter number of units 500
Total Bill Amount 3500
```

Program 3: Orange Level – Employee Salary Breakup

Problem Statement:

An organization calculates employee salary components from basic pay.

Write a C program that reads Employee ID and Basic salary

Calculate and display: House Rent Allowance (HRA) – 20% of basic, Dearness Allowance

(DA) – 15% of basic, Gross salary

INPUT :

```c
#include<stdio.h>
void main()
{
int id;
float basic,hra,da,gross;
printf("Enter the employee id and basic salary: ");
scanf("%d",&id);
scanf("%f",&basic);
hra=0.2*basic;
da=0.15*basic;
gross=hra+da+basic;
printf("HRA : %f\nDA: %f\nGross : %f",hra,da,gross);
}
```

OUTPUT :

Output

```
Enter the employee id and basic salary: 23 500000
HRA : 100000.000000
DA: 75000.000000
Gross : 675000.000000
```

Program 4: Orange Level

Write a program for a coffee shop that calculates the total cost of an order. Take the number

of coffees and price per coffee as input, then apply a 5% service charge. Display the final

amount with 2 decimal places.

INPUT :

```c
#include<stdio.h>
#define service 0.05
void main()
{
float n,pr, final;
printf("Enter no of coffees and price per coffee: ");
scanf("%f %f",&n,&pr);
final= n*pr+service;
printf("Final price = %.2f", final);
}
```

OUTPUT :

```
Enter no of coffees and price per coffee: 5 500
Final price = 2500.05
```

Program 5: Banana Level - Vehicle Fuel Cost Calculator

Problem Statement:

A travel agency estimates fuel expenses for a trip.

Write a C program that reads Total distance to travel (in km), Mileage of the vehicle (km per

liter), Cost of fuel per liter

The program should calculate and display: Fuel required for the trip, Total fuel cost

INPUT :

```c
#include<stdio.h>
void main()
{
int TD,Mileage,Cost;
float Fuel_req,Total_Cost;
printf("Enter the total distance of the trip (in kms) : ");
scanf("%d",&TD);
printf("Enter the Mileage od the Vehicle (km/l): ");
scanf("%d",&Mileage);
printf("Enter the cost of fuel per liter: ");
scanf("%d",&Cost);
Fuel_req=TD/Mileage;
Total_Cost=Fuel_req*Cost;
printf("the Fuel required for the trip = %.2f\n",Fuel_req);
printf("The total cost required for the trip =%.2f\n",Total_Cost);
}
```

OUTPUT :

```
Enter the total distance of the trip (in kms) : 50
Enter the Mileage od the Vehicle (km/l): 25
Enter the cost of fuel per liter: 102
the Fuel required for the trip = 2.00
The total cost required for the trip =204.00
```

Program 6: Banana Level - Simple Bank Account Transaction

Problem Statement:

A bank wants to simulate a simple account transaction.

Write a C program that reads Initial account balance, Amount deposited, Amount withdrawn

The program should calculate and display: Final account balance

Assume withdrawal amount is always less than balance.

INPUT :

```c
#include <stdio.h>
int main()
{
float initialBalance, deposit, withdraw, finalBalance;
printf("Enter initial account balance: ");
scanf("%f", &initialBalance);
printf("Enter amount deposited: ");
scanf("%f", &deposit);
printf("Enter amount withdrawn: ");
scanf("%f", &withdraw);
finalBalance = initialBalance + deposit - withdraw;
printf("Final account balance = %.2f\n", finalBalance);
return 0;
}
```

OUTPUT :

```
Enter initial account balance: 50000
Enter amount deposited: 830
Enter amount withdrawn: 10000
Final account balance = 40830.00


=== Code Execution Successful ===
```

Program 7: Orange Level

Problem Statement:

Write a program to calculate the volume of a cone using: $V = (1/3)\pi r^2 h$, where r is radius and h is height. Use #define for π and display with 3 decimal places.

INPUT :

```c
#include <stdio.h>
#define pi 3.14
int main()
{
float r,h,volume;
printf("Enter the value of radius and height : ");
scanf("%f %f",&r,&h);
volume=0.333*pi*r*r*h;
printf("The Volume of Cone is : %.3f", volume);
return 0;
}
```

OUTPUT :

```
Output

Enter the value of radius and height : 7 20
The Volume of Cone is : 1024.708

=== Code Execution Successful ===
```

Program 8 : Orange Level

Problem Statement:

Write a C program to read the temperature in Celsius and convert it into Fahrenheit using

the formula and display the converted temperature.

$$F = (C \times 9/5) + 32$$

INPUT :

```c
#include <stdio.h>
int main()
{
float f,c;
printf("Enter the temp in Celcius : ");
scanf("%f",&c);
f=(c*(9.0/5.0))+32;
printf("The temp in Fahrenheit is : %f",f);
return 0;
}
```

OUTPUT :

Output

```
Enter the temp in Celcius : 30
The temp in Fahrenheit is : 86.000000

=== Code Execution Successful ===
```